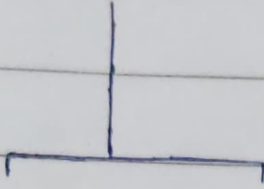


Frame: A frame is a coordinate system.



Inertial frame

Non-inertial frame.

(either at rest or

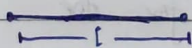
(accelerated frame)

moving with constant ~~acceleration~~

velocity)

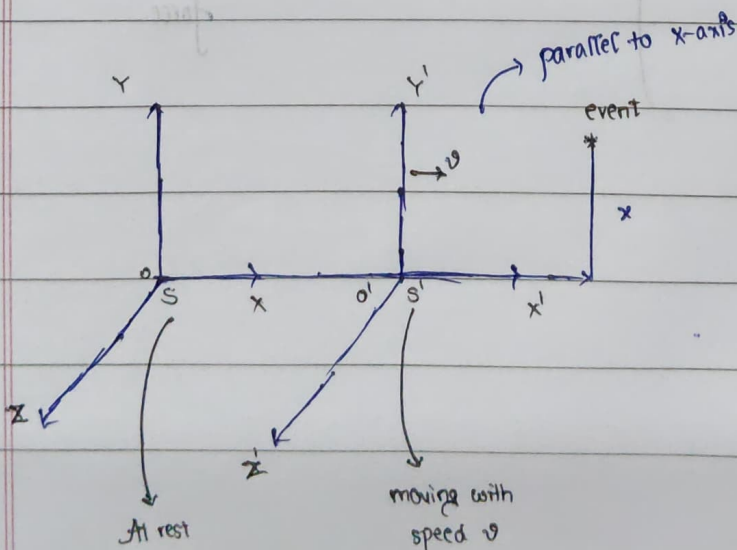
Relativity - Information shared between observers, one is at rest and another is moving with constant velocity (both are in inertial frame).

The fundamental quantities mass, length, time remain invariant in our day to day life.



Two observers will measure the length of rod to

be same irrespective of location.

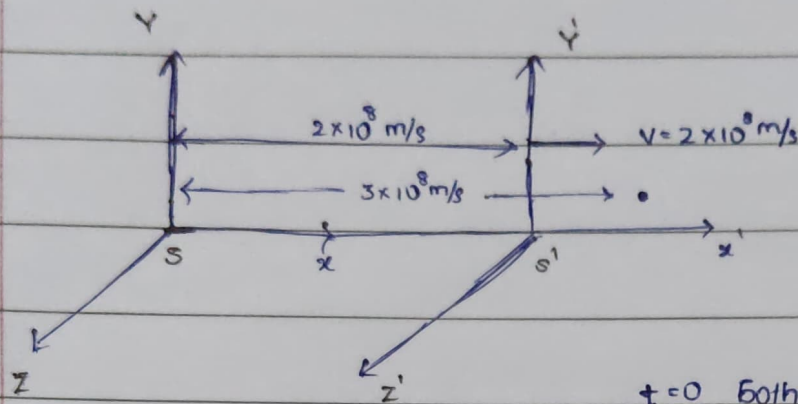


Objective: To describe the coordinates of the ~~same~~ event by the two observers S and S' . ($v \ll c$)

$x' = x - vt$	Galilean Coordinate Transformation	$v'_x = v_x - v$	Galilean Velocity Transformation
$y' = y$		$v'_y = v_y$	
$z' = z$		$v'_z = v_z$	
$t' = t$			

Take further derivatives

$a'_x = a_x$	Galilean Transformation for the acceleration and force.
$a'_y = a_y$	
$a'_z = a_z$	
$f'_x = f_x$	
$f'_y = f_y$	
$f'_z = f_z$	



$t=0$ Both S and S' coincide with each other.

After $t=1$ for S $3 \times 10^8 \text{ m/s}$
for S' $1 \times 10^8 \text{ m/s}$.

Einstein Postulates :

- (1) Physics principles have to be same in all inertial frame of references
- (2) The speed of light is same in all inertial frame of references.